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Dynamic capabilities for omnichannel transformation in MSMEs: A comparative case study of fashion and furniture sectors

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ABSTRACT

Omnichannel is one of the most recent trends in online shopping, retail, and contact centers. It provides a seamless and integrated customer experience across all available communication channels. MSMEs play a crucial role in developing economies, but their journey towards omnichannel integration faces significant challenges due to limited resources, managerial commitment, and technological adaptability. This study explores the stages of transformation towards omnichannel adoption among micro, small, and medium enterprises (MSMEs) in fashion and furniture, as representative subsectors of Indonesia's creative economy, focusing on the dynamic capabilities. This research addresses the gap in understanding how different MSMEs navigate these challenges through various stages of transformation. Using a multiple case study approach, the research investigates two MSMEs in fashion and furniture industries through in-depth interviews and document analysis. Findings reveal that fashion MSMEs leverage real-time inventory management and personalized customer engagement for rapid integration, whereas furniture MSMEs adopt incremental approaches focused on logistical efficiency and collaboration with external partners. By emphasizing sector-specific applications of dynamic capabilities—sensing, seizing, and transforming—this study underscores the importance of collaborative alliances and leadership in addressing operational challenges. Practical implications include strategies for MSMEs to enhance competitiveness through tailored omnichannel solutions. These insights offer practical implications for MSMEs aiming to enhance their competitive edge through omnichannel strategies.

1. Introduction

Micro, small and medium enterprises (MSMEs) are vital to most economies, especially in developing countries. MSMEs are the backbone of the global economy, driving job growth, competitiveness and innovation as well as Gross Domestic Product (GDP) (Sopha and Hestiani, 2018). MSMEs serve as a platform for improving entrepreneurship and business skills, and they can employ numerous workers to contribute to the growth of the rural economy. In Indonesia, the fashion and furniture industries stand out as critical subsectors within the creative economy. These industries are defined by their emphasis on creativity, skills, and talent to produce both economic and cultural value. Within the MSME landscape, fashion and furniture enterprises contribute significantly to the nation's GDP and highlight the diversity of creative economic activities (Ministry of Tourism and Creative Economy, 2020). This study focuses on these subsectors to explore how dynamic capabilities enable their transformation toward omnichannel strategies. However, its development was hampered by several obstacles, including debt payments, employee salary payments, bill payments, production raw material costs, rent payments, capital reduction, customer attrition, and most importantly, difficulties in marketing and sales (Adivar et al., 2019). These constraints can differ from one region to another, it can be between different sectors, or between actors within a sector (Ailawadi and Farris, 2017).

Omnichannel transformation has become a critical strategy for MSMEs to remain competitive, particularly in an increasingly digitalized market where customer expectations demand seamless integration across channels. Technological developments have affected changes in consumer behavior. Consumers now expect convenience and speed in their purchasing process, utilizing various digital platforms to reach their needs instantly (Kannan and Li, 2017). This customer behavior encourages companies, including MSMEs, to evaluate their business models to remain relevant and competitive in a dynamic market. In an increasingly competitive environment, MSMEs are faced with the

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challenge of not only surviving local but also global competition, which encourages them to adopt more flexible and responsive business models such as omnichannel. Omnichannel can be defined as the integration of multiple channels allowing customers to make purchases through any channel they choose (Verhoef et al., 2015). Additionally, omnichannel also offers a consistent and optimal customer experience through the accessibility of more information (Saghiri et al., 2017). In practice, omnichannel allows customers to access, purchase, and receive products through all available channels in a fully integrated manner (Kumar and Uma, 2019; Verhoef et al., 2015). Omnichannel implementation faces several obstacles that not only involve technology implementation, but also business transformation that affects customer approaches and business models (Bell et al., 2014; de Borba et al., 2021; Hoogveld and Koster, 2016). Dynamic capabilities, which emphasize sensing, seizing, and transforming opportunities, provide a robust framework for understanding the complex transformation processes required for successful omnichannel implementation (Teece, 2007).

This research addresses the critical gap in existing literature, particularly the lack of understanding of how dynamic capabilities operate across sectors during omnichannel transformation. While prior studies (Barbosa dan Casais, 2022) have explored the general role of dynamic capabilities, the present research highlights sector-specific applications, especially in MSMEs with resource constraints. By examining both fashion and furniture MSMEs, this study highlights how the unique characteristics of these sectors influence the deployment of dynamic capabilities. For instance, the fashion sector emphasizes rapid inventory turnover and customer interaction, whereas the furniture sector prioritizes logistics optimization and material procurement. These insights contribute to a more nuanced understanding of omnichannel adoption in MSMEs and advance the theoretical framework of dynamic capabilities in omnichannel transformation.

MSMEs are currently using online channels like websites, e-commerce, and social media to complement their offline channels (physical stores). It is common to find that the integration of these channels is only partially. The gap is a consequence of a shortage of certain abilities necessary for a successful transition to an omnichannel approach. An organization's ability to adapt and transform to take advantage of opportunities in an ever-changing market is called dynamic capabilities (Teece, 2007). Dynamic capabilities provides a framework for managing this complexity, enabling MSMEs to integrate and coordinate cross-functional operations more effectively (Eisenhardt and Martin, 2000). However, limited research has explored the specific dynamic capabilities required by MSMEs to navigate these transformations. To address this gap, this study seeks to answer the following research questions:

RQ 1 How do dynamic capabilities shape the stages of transformation that MSMEs experience during the implementation of omnichannel?

RQ 2 What dynamic capabilities are required by MSMEs to address challenges and opportunities during their omnichannel transformation?

The findings of this research reveal that MSMEs with stronger resources and strategic vision progress more swiftly towards full omnichannel integration, while others adopt gradual approaches due to resource constraints. By exploring these dynamics, this study provides a nuanced understanding of omnichannel adoption in MSMEs, contributing both theoretical insights and practical implications.

This paper contributes novel insights into the sector-specific nature of omnichannel transformation, offering practical implications for resource-constrained MSMEs and theoretical advancements in the application of dynamic capabilities. The structure of this paper is as follows. Section 2 discusses the theory on which this research is based. Section 3 explains the research method, research design and data collection. Section 4 presents the research findings and analysis of the semi-structured interviews as well as the results of the observations made. Finally, summarized conclusions along with research limitations and future research are presented in Section 5.

2. Literature review

2.1. Omnichannel transformation

The development of sales channel concepts has evolved from single channel, multichannel, to omnichannel, marked by increasingly higher levels of integration (Ailawadi and Farris, 2017; Beck and Rygl, 2015). Omnichannel is defined as a fully integrated approach (Brynjolfsson et al., 2013; Verhoef et al., 2015; Xu and Cao, 2019). Cao and Li (2015) conducted a comparative study of 71 US retailers, identifying several stages of cross-channel integration strategies. Integration starts from multichannel - silo mode (level 1), when online and traditional are operated separately, multichannel - minimal integration (level 2), then multichannel - moderate integration (level 3), and the last stage is omnichannel - full integration (level 4). Although this model has been widely accepted, Barbosa and Casais (2022) highlight that its applicability in MSMEs might require adjustments to account for resource constraints and sector-specific dynamics.

While many studies assume a sequential progression through these stages, other research suggests that companies with strong resources and dynamic capabilities can skip stages or directly achieve full integration. For example, Berman and Thelen (2016) emphasize that companies with adequate resources and strong managerial commitment may bypass intermediate stages. Similarly, Simone and Sabbadin (2018) highlight that firms with mature digital strategies and a strong understanding of consumer behavior can achieve omnichannel integration without following every stage. These findings underscore the importance of contextual factors, such as company size, market served, and technology readiness, which shape the transformation paths firms take. Based on the aforementioned literature, the transformation shifting from multi-channel to omnichannel is shown in Fig. 1.

Fig. 1 illustrates the transformation process from multi-channel to omnichannel, adapted and synthesized from Cao and Li (2015). This framework highlights the progressive integration of channels, which forms the basis for analyzing the stages of MSMEs transformation in this study. Arrows with different colours indicate potential transformation paths as evidenced by Berman and Thelen (2016) and Simone and Sabbadin (2018).

Omnichannel aims to combine all available touchpoints, resulting in increased sales that can be achieved through increased customer interaction. The focus is primarily on customers and integration (Larke et al., 2018). The implementation of omnichannel requires managers to create new marketing policies and strategies that consolidate interactive retail channels and traditional mass advertising channels. Combining traditional and modern communication channels will allow for an expanding consumer base (Park and Lee, 2017). Providing a seamless customer experience is a crucial aspect for consumers in omnichannel. After trying products in the store, consumers have the option to purchase them online. Consumers who have searched for a desired product online can purchase it from a physical store at the same time (Verhoef et al., 2015). Despite its advantages, omnichannel implementation poses significant challenges. Integrating data across channels and ensuring its security is one of the most fundamental issues (Chen et al., 2018). In addition, channel integration often requires high investment costs (Saghiri and Mirzabeiki, 2021). Previous studies have primarily concentrated on larger organizations, leaving a gap in understanding how MSMEs manage omnichannel transformations through dynamic capabilities. Due to their limited resources, MSMEs encounter greater challenges in this process, making dynamic capabilities even more essential for successfully navigating the transformation.

2.2. Dynamic capabilities in omnichannel

Omnichannel implementation has significant implications for customer interactions and transactions, as well as company resources and capabilities (Lehrer and Trenz, 2022). Dynamic capabilities refer to

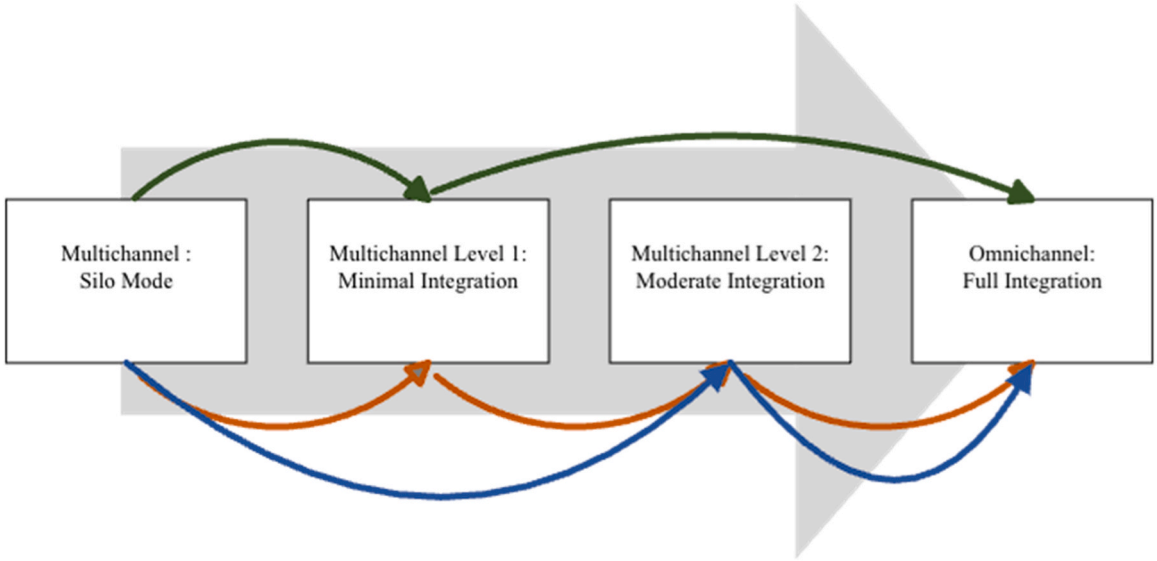


Fig. 1. Transformation from multi-channel to omnichannel (Adapted from Cao and Li, 2015).

an organization's ability to continuously adapt, build, and transform its resource base in response to changes in a dynamic environment by (Teece et al., 1997). The implementation of omnichannel requires both routine and dynamic capabilities, including integrated software systems that consolidate product, customer, shopping, and communication data, making them accessible across both online and physical channels (Beck and Rygl, 2015). Solem et al. (2023) identified key dynamic capabilities required for omnichannel transformation, including the optimization of digital ERP systems, the ability to integrate marketing communications, and collaboration across business units and with external partners. These capabilities are critical for MSMEs, as they enable efficient resource management and foster innovative solutions despite resource constraints.

In the digital and omnichannel era, dynamic capabilities are key factors for organizations to stay relevant and competitive. Dynamic capabilities include the ability to detect changes in the market (sensing), capture opportunities (seizing), and change and reconfigure organizational assets (transforming) to create sustainable competitive advantage (Mrutzek-Hartmann et al., 2022a,b). Sensing is about gathering information and recognizing opportunities or threats in the environment. Seizing referring to an organization capacity to capitalize on opportunities in a rapidly evolving environment. Transforming involves an organization's ability to continuously adapt and innovate its resources and processes to stay competitive. The dynamic capabilities framework that guides the research is shown in Fig. 2.

Mrutzek-Hartmann et al. (2020) conducted a content-based literature analysis of 63 academic papers identified by a Scopus literature search. It stated that there are six constructs of dynamic capabilities in omnichannel, namely customer/market understanding, integration and communication, customer interaction, omnichannel environment, supply chain management as well as innovativeness. This study divides dynamic capabilities into front-end and back-end activities, aiming to separate processes that directly interact with customers from internal operations. Front-end activities, such as customer interaction and market understanding, focus on creating a smoother customer experience across multiple channels. This is important for enhancing satisfaction and delivering personalized experiences. Meanwhile, back-end activities, such as supply chain management and integration, focus on internal operations like logistics and inventory management. These activities support the frontend processes by ensuring that operations run efficiently.

Building on this, Mrutzek-Hartmann et al. (2022a,b), conducted a study on omnichannel resources and capabilities of MSME retailers in Germany and Turkey. The study emphasized the importance of dynamic capabilities, such as customer and market insights, channel integration, and supply chain management, which are critical to the success of an

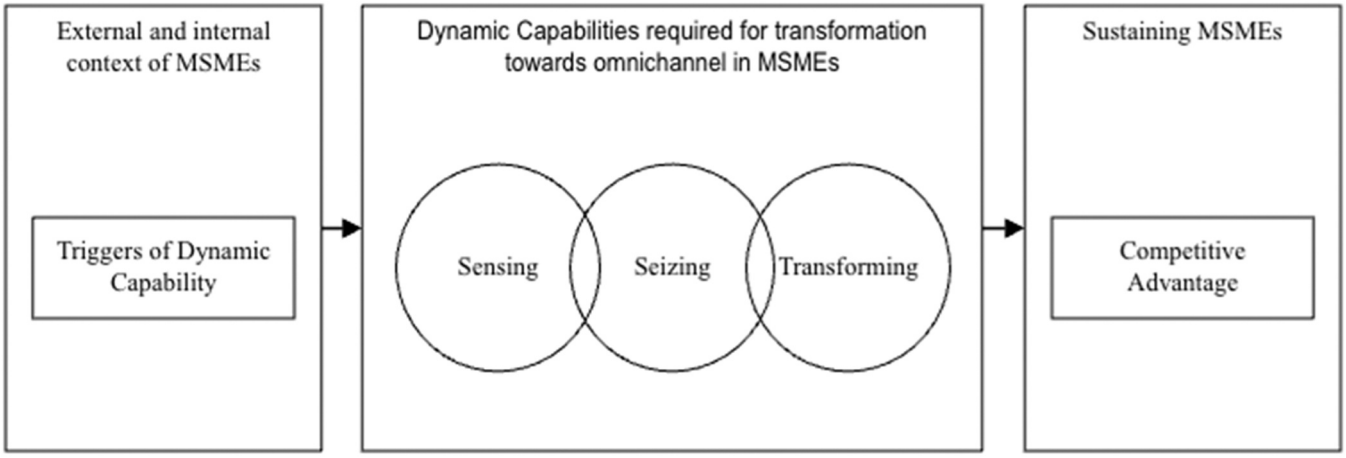


Fig. 2. Dynamic capabilities framework.

omnichannel implementation. Moreover, significant differences were observed across countries, particularly in the allocation and application of resources and capabilities. This indicates that contextual factors, such as market maturity and technology readiness, play a significant role in shaping the success of omnichannel strategies. For MSMEs, this highlights the importance of leveraging dynamic capabilities to overcome structural and operational limitations, ensuring competitiveness in increasingly complex markets.

Omnichannel implementation is not just about integrating various sales channels, but also involves complex structural changes and adaptive resource management. In this process, dynamic capabilities become crucial because they enable companies to manage operational complexity, innovate in customer service, and adapt quickly to market changes (Barbosa and Casais, 2022; Cao and Li, 2015; Hansen and Sia, 2015). For MSMEs, these capabilities are particularly critical given their resource constraints and the need for agile adaptation to customer demands. Cao and Li (2015) explained that the transformation to omnichannel often goes through several stages of integration, but not all companies have to follow this sequence linearly. Companies with strong dynamic capabilities are able to skip certain stages and move directly to a higher level of integration. This emphasizes that dynamic capabilities, such as fast decision making and technology adaptation, are key drivers in accelerating the transformation to omnichannel (Berman and Thelen, 2016; Simone and Sabbadin, 2018).

Hansen and Sia (2015) demonstrate how dynamic capabilities can be used by organizations to effectively tackle challenges in channel integration, such as data synchronization, consistent customer experiences, and team coordination. Despite these insights, the specific dynamic capabilities required for MSMEs in resource-constrained settings remain underexplored, forming a critical research gap. Key aspects, such as how MSMEs can effectively leverage digital tools, foster cross-channel collaboration, and enhance workforce adaptability with limited resources, are not well-documented. This study addresses these gaps by examining sector-specific strategies and practical applications of dynamic capabilities to support MSME omnichannel transformations in constrained environments. For MSMEs, overcoming logistical inefficiencies and aligning internal operations with customer expectations is essential despite limited resources.

The successful execution of omnichannel strategies necessitates organizational changes that foster collaboration among departments, encouraging interaction and mitigating silos, and ensuring alignment between various operational processes. This includes seamless communication, synchronized data sharing, and cohesive efforts across marketing, inventory, and customer service functions to deliver a unified customer experience (Barbosa and Casais, 2022; Piotrowicz and Cuthbertson, 2014). Dynamic capabilities are central to this process, as they enable firms to reconfigure resource bases, integrate processes, and maintain a customer-centric approach. For MSMEs, this includes reallocating resources efficiently, adapting to new technologies, and enhancing coordination among limited staff and infrastructure. Nevertheless, critical factors essential for effective omnichannel integration remain inadequately understood, particularly in the context of MSMEs (Galipoglu et al., 2018).

Barbosa and Casais (2022) provided valuable insights into how dynamic capabilities underpin the successful implementation of omnichannel strategies. However, these insights primarily focus on larger organizations, leaving a gap in understanding their application within MSMEs. By analyzing how MSMEs leverage sensing, seizing, and transforming capabilities, this research sheds light on their ability to overcome structural and operational challenges. For instance, sensing capabilities in MSMEs enable them to identify shifts in consumer behavior and emerging market opportunities, even with limited access to advanced analytics tools. Seizing capabilities, on the other hand, are reflected in the strategic allocation of minimal resources to achieve maximum impact, such as prioritizing digital marketing channels that offer the highest return on investment. Finally, transforming capabilities empower MSMEs to adapt their operational models incrementally, allowing for flexibility and resilience despite resource constraints.

This research bridges the gap identified in Barbosa and Casais (2022) by providing empirical evidence on the specific dynamic capabilities required for MSMEs. While previous studies emphasized the theoretical significance of these capabilities, this study offers practical insights into their application. By focusing on MSMEs with varying levels of digital maturity, the findings highlight tailored approaches that align with their unique constraints and opportunities. These insights contribute to a more comprehensive understanding of how dynamic capabilities facilitate omnichannel transformation in diverse MSME contexts.

2.3. Dynamic capabilities and open innovation

Dynamic capabilities and open innovation share a synergistic relationship in enabling firms, particularly MSMEs, to adapt and thrive in dynamic environments. Collaborative alliances, a core element of open innovation, play a pivotal role in extending the resource base of MSMEs, allowing them to access external expertise, networks, and innovation opportunities beyond their internal capabilities (Chesbrough, 2003). These alliances not only mitigate resource constraints but also foster co-creation and the development of innovative solutions tailored to evolving market demands. Similarly, leadership acts as a critical enabler of open innovation by embedding a culture of experimentation, collaboration, and agility within the organization (Chesbrough and Bogers, 2014). Effective leaders align innovation processes with organizational goals, ensuring seamless integration of internal and external innovation efforts while promoting strategic decision-making to capitalize on emerging opportunities. Together, collaborative alliances and leadership amplify the adaptive potential of dynamic capabilities, highlighting their critical role in driving MSMEs' transformation towards omnichannel strategies in resource-constrained settings.

2.4. Dynamic capabilities-based propositions for MSMEs' omnichannel transformation

Based on the literature review and the key dynamic capabilities identified, this study formulates the following propositions to guide the analysis of MSMEs' transformation towards omnichannel implementation. These propositions aim to reflect the operational realities and challenges faced by MSMEs, providing a conceptual lens for the subsequent qualitative analysis.

1. Proposition 1 MSMEs with resource constraints tend to prioritize operational stability and incremental progress over comprehensive and rapid omnichannel transformation.
2. Proposition 2 Dynamic capabilities, such as customer interaction and market understanding, influence the speed and effectiveness of omnichannel transformation based on the specific operational challenges faced by each MSME.
3. Proposition 3 Variations in organizational focus, such as prioritizing inventory accuracy or logistics optimization, shape how MSMEs develop and leverage dynamic capabilities during omnichannel transformation.
4. Proposition 4 Collaborative alliances, leadership, and trust are critical enablers for MSMEs in overcoming resource limitations and fostering successful omnichannel integration.

These propositions guide the design of the qualitative study, focusing on how MSMEs utilize dynamic capabilities to navigate the complexities of omnichannel transformation. The methodology section elaborates on how these propositions are explored through case studies.

3. Methodology

This research uses a qualitative approach using a case study method

to gain an in-depth understanding of the MSME transformation process towards omnichannel. A case study is a research methodology that emphasizes the comprehension of dynamics occurring within a specific context or environment (Eisenhardt, 1989). Case studies can encompass either individual or multiple cases and may include various levels of analysis, hence generally integrate multiple data collection methods, including archives, interviews, questionnaires, and observations (Yin, 2009). The evidence gathered can be qualitative (e.g., words), quantitative (e.g., numbers), or a combination of both (Eisenhardt, 1989).

This study employs a case study method because it provides a deep understanding of complex phenomena, such as the omnichannel transformation in MSMEs. This method allows for a comprehensive exploration of various variables involved, such as dynamic capabilities, technology integration, and operational adaptation, which are difficult to capture through quantitative methods. Furthermore, the case study is relevant in the context of MSMEs, which have specific characteristics, including resource limitations and unique market environments. By utilizing multiple data sources such as interviews, observations, and documents, this method ensures strong data triangulation, enhancing the validity of the research findings. Additionally, the case study method allows for comparisons between sectors, such as fashion and furniture, providing broader insights and stronger generalization regarding the role of dynamic capabilities in the omnichannel transformation process. The research design using the case study approach in this study is presented in Table 1.

3.1. Getting started

This study is aimed at exploring the omnichannel implementation process for MSMEs, identifying the stages of transformation they go through, and examining the dynamic capabilities needed for this transition. The dynamic capabilities framework is used to analyse the transformation process of MSMEs towards omnichannel implementation. Dynamic capabilities refer to an organization ability to sense opportunities and threats, seize those opportunities, and undertake the necessary transformations to maintain competitiveness in an ever-changing market. Three main aspects are examined in this context: sensing, seizing, and transforming (Teece, 2007).

Table 1
The research design using the case study approach.

Step	Activity
Getting started	Developing the research question, setting the study's objectives
Selecting case	Selecting the cases to be studied based on criteria that match the research question
Crafting instruments and protocols	Designing and preparing the tools for data collection. Developing interview guides, surveys, and observation sheets
Entering the field	Simultaneous data collection and analysis, by: Initiating contact with the selected cases, arranging interviews, and beginning data collection Data collection and analysis are conducted simultaneously, allowing for ongoing adjustments and refinements as insights emerge during the research process.
Analysing	Conducting cross-case analysis using coding and thematic analysis to identify patterns, categorizing the data, and drawing conclusions
Shaping propositions	Refining and developing initial propositions based on preliminary insights, observations, or data, by: Iterative tabulation of evidence for each construct Replication, not sampling, logic across cases Search evidence for 'why' behind relationship
Enfolding literature	Comparing findings with existing literature, identifying gaps, and supporting the analysis with theoretical frameworks
Reaching closure	Ensuring theoretical saturation, finalizing the analysis, and preparing conclusions

3.2. Selecting case

The focus of this research is on MSMEs that operate in the creative economy sector in Indonesia. These sectors were chosen due to their significant contribution to the national economy and their potential to leverage omnichannel strategies to enhance competitiveness. The primary factors in choosing the case of this research are MSMEs that manufacture goods and then sell them to consumers directly, both online and offline. Furthermore, these MSMEs should have at least national market coverage to make them relevant to the research topic. MSMEs must undergo a transformation process towards omnichannel to meet the final criteria. This criterion is based on the findings of previous research that indicate successful implementation of omnichannel is marked by the elimination of differences between traditional and online channels (Melacini et al., 2018; Verhoef et al., 2015; Wollenburg et al., 2018).

The selected cases enable to record the transformation towards omnichannel, from zero integration to full integration. In addition, case selection allows comparisons between MSMEs as they transition through these stages. This approach and criteria led to the acquisition of two MSMEs as research objects. The first MSME is an MSME that operates in the fashion sector with three physical stores, actively participating in bazaars in various cities, as well as using social media, websites and e-commerce platforms as sales channels. The second MSME is an MSME that operates in the furniture sector with two physical stores as offline sales channels, utilizing social media and e-commerce platforms as online sales channels. Table 2 presents the profile of MSMEs that meet the case criteria.

3.3. Crafting instruments and protocols

Developing an interview guide and observation sheet is an important aspect of this research, ensuring that the data collected was comprehensive and relevant to the research objectives. The interview guide was designed using the dynamic capabilities theoretical framework as presented in Section 2.2 to facilitate semi-structured interviews with MSME owners and employees. The interview guide included questions that encouraged participants to share their views on strategic insights and daily operational experiences in the omnichannel transformation. Observation sheets are also created to systematically record field observations, adding context to the interview results. These sheets are designed to track important operational changes and their effects on the business, ensuring data collection is consistent and objective.

To ensure alignment between the research objectives, the theoretical framework, and data collection process, operational definitions of key variables and themes were established. These definitions provided a foundation for crafting the interview guide and observation sheets, ensuring that each question and observation criterion was directly linked to the study's focus. Table 3 presents the operational definitions of the key variables used in this research.

Combined with the interview guides, these tools help gather detailed, diverse data to better understand the omnichannel transformation in MSMEs. The interview guide can be found in Appendix 1.

3.4. Entering the field

The main data source used in this research is the results of semi-structured interviews, then supplemented with secondary data in the form of MSMEs reports, data from social media, websites, and notes from field observations. The combination of data collection will enrich the information (Eisenhardt, 1989). The interview subjects are the owners and several employees of two MSMEs, with the objective of examining the diverse viewpoints of each individual interviewed (Eisenhardt and Graebner, 2007). MSME owners were selected as

Table 2
The profile of MSMEs.

MSME	Business Type	Years of operation	Number of Employee	Offline sales channels	Online sales channels	Characteristics
Fashion	Ready-to-wear fashion	11 years	15 persons	3 physical stores and bazaar	Social media, marketplace and website	Emphasized real-time inventory updates and customer engagement as central to their omnichannel strategy. Prioritized brand presence across digital channels, using platforms like Instagram to enhance customer interaction and loyalty.
Furniture	Custom wooden furniture	11 years	7 persons	2 physical stores	Social media, marketplace	Focused on logistical coordination and raw material procurement to meet bulk orders and sustain production schedules. Reliance on collaborative partnerships and local networks provided resilience despite technological constraintsPrioritized stability, rather than rapid digital adoption to maintain business continuity.

Table 3
Operational definitions of the variables.

Variable/Theme	Operational Definition	Reference
Sensing	The ability of MSMEs to identify customer needs, market opportunities, and potential threats through data collection, analysis, and feedback.	Teece et al. (1997)
Seizing	The capacity of MSMEs to mobilize resources, develop strategies, and capitalize on identified opportunities to address customer demands and market changes.	Eisenhardt and Martin (2000); Teece et al., (1997)
Transforming	The ability of MSMEs to adapt, restructure, and innovate internal processes, resources, and business models to sustain competitiveness and ensure omnichannel integration.	Teece et al. (1997)
Customer (market) understanding	The ability of MSMEs to analyze customer needs, preferences, and market trends to adapt products and services accordingly.	Mrutzek-Hartmann et al. (2020)
Integration and coordination	The processes and systems used to synchronize and align operations, inventory, and information across online and offline channels.	Cao and Li (2015); Mrutzek-Hartmann et al. (2020)
Customer Interaction	The methods and tools used to engage with customers across multiple touchpoints, ensuring seamless and consistent experiences.	Mrutzek-Hartmann et al. (2020); Solem et al. (2023)
Omnichannel Environment	The infrastructure, resources, and systems that enable MSMEs to operate seamlessly across physical and digital channels.	Mrutzek-Hartmann et al. ((2020); Saghiri and Mirzabeiki (2021)
Innovativeness	The capacity of MSMEs to develop and implement creative strategies, technologies, or processes to improve their competitiveness.	Mrutzek-Hartmann et al., (2020)

interviewees because they possess strategic insight and decision-making authority to assist in the shift to omnichannel. Meanwhile, employees were selected because they run daily operations and have direct experience with the changes that occur, so they can provide in-depth insight into how these changes affect operational effectiveness. By selecting owners and employees as interviewees, this research can collect data covering various aspects and perspectives of the omnichannel transformation process, thereby providing deeper and more comprehensive insights for the improvement and success of the strategy adopted. Table 4 provides an overview of the interviewee profiles, including their roles within the organization, years of experience, and their relevance to the study.

Interviews were conducted by asking open-ended questions. To give more in-depth insight, each interviewee is sent these questions prior to their interview schedule. The length of interviews ranged from 30 to 120 min, recorded and transcribed for subsequent analysis. Each MSME agreed to conduct an in-office interview in January - March 2024. Data collection and analysis are conducted simultaneously, allowing for ongoing adjustments and refinements as insights emerge during the research process. The interviews were conducted in Indonesian, and the statements that were used then translated into English and checked for linguistic equivalence, quality and plausibility using a back translation procedure. Interview protocols with leading questions are very effective in helping researchers explore topics in depth. In this approach, the researcher has a planned framework of questions but still has the flexibility to dig deeper based on the interviewee's responses. This makes it possible to follow up on interesting and unexpected responses, while interviewees are free to elaborate on their perceptions, experiences and reflections (Paré, 2004).

The interview processes consisted of two rounds. The first round aimed to gather initial insights from the respondents, while the second round was conducted to strengthen and validate the preliminary findings. The second round of interviews revealed no new information or insights, indicating that data saturation had been achieved. This confirmed that the data collected was sufficient, and no additional respondents were needed. Saturation occurs when the information provided by respondents becomes repetitive, with no further variations or novel insights emerging (Denzin, 1988). In addition, secondary data collected is in the form of MSME information available on social media, websites, e-commerce platforms, observation notes and internal documents (consumer data, sales data, organizational structure, minutes of meeting, etc.). These documents can provide additional information regarding the transformation process and strategy for each MSME. The unit of analysis in this research includes the organization, its dynamics and internal activities.

3.5. *Analysing*

Thematic analysis (Braun and Clarke, 2019) was conducted to identify patterns and themes across the cases. The analysis focused on how MSMEs leveraged their dynamic capabilities—sensing, seizing, and transforming—to address sector-specific challenges and implement omnichannel strategies. This process began with data coding, where key concepts and recurring elements were tagged and categorized using NVIVO software. NVIVO facilitates the organization of large datasets by allowing researchers to efficiently code and manage qualitative data. Through thematic analysis, these codes were grouped into broader themes, revealing underlying patterns and relationships across the cases studied. By categorizing the data in this way, researchers were able to draw meaningful conclusions, highlighting similarities and differences that inform the overall findings of the study and contribute to a deeper understanding of the research questions. Fig. 3 illustrates the data structure derived from the content analysis process, showing the transition from open codes to categorized codes, and finally to thematic

Table 4
The profile of interviewees.

MSMEs	Interviewee Role	Years of Experience	Key Contribution
Fashion	Owner (speaker 1)	11	Providing strategic insights into branding, marketing, and product development while highlighting the challenges and opportunities of integrating offline and online channels for a seamless omnichannel experience.
	Creative and marketing director (speaker 2)	11	Enhancing operational efficiency by managing daily operations, overseeing social media engagement, and improving customer experiences through effective communication and issue resolution strategies.
Furniture	Owner (speaker 1)	11	Leading product diversification and supply chain development, sharing strategic insights on balancing customer satisfaction and operational costs in omnichannel integration.
	Co-owner and marketing leader (speaker 2)	10	Managing promotional activities and customer engagement on social media and marketplaces, highlighting the challenges of competition and emphasizing the importance of innovation in product offerings.
	Supervisor (speaker 3)	9	Supporting order and inventory management, ensuring smooth customer interactions, and maintaining consistency across e-commerce platforms, as well as, contributing to data organization for operational decision-making.

findings. This step-by-step progression captures how raw qualitative data were transformed into meaningful insights, which are directly aligned with the research questions.

The content analysis yielded 261 open codes from diverse data sources, such as interview transcripts, social media content, and field notes. These open codes were grouped into 41 broader categories through an iterative process of comparison and refinement. The final thematic groups represent the essence of these categories and include constructs like 'dynamic capabilities,' 'customer engagement,' and 'technology integration.' To ensure consistency and accuracy, the themes were reviewed and validated against relevant literature and theory (Braun and Clarke, 2019).

The cross-case analysis involved comparing the thematic findings across the two MSMEs to identify similarities and differences in their omnichannel transformation processes. For instance, while both MSMEs highlighted challenges in inventory management, fashion MSMEs emphasized real-time data synchronization, whereas furniture MSMEs focused on logistics and material procurement. This approach provided a nuanced understanding of how dynamic capabilities manifest differently in each sector (Yin, 2009).

3.6. Shaping propositions

Propositions were formulated based on the thematic analysis results. This iterative process began with coding the interview data, followed by categorizing these codes into broader themes. These themes were then aligned with the theoretical framework to derive propositions that encapsulate the key insights. This approach ensures that the propositions are reflective of both the empirical data and the theoretical lens used in the study (Braun and Clarke, 2019; Creswell and Creswell, 2017).

3.7. Enfolding literature

In this research, enfolding literature means comparing the findings from the case studies of MSMEs in the fashion and furniture sectors with existing theories on omnichannel implementation and dynamic capabilities. This comparison helps identify gaps between theory and what was observed. By aligning the study's results with established theories, the research can draw stronger conclusions and highlight where current theories might need updating.

3.8. Reaching closure

As the study progresses, theoretical saturation is achieved when no new insights or patterns emerge from the data, indicating that the key themes and constructs have been fully explored. At this point, the analysis is refined and conclusions are drawn based on the comprehensive understanding gained from the data. The use of NVIVO software facilitates this process by organizing and categorizing the data, allowing the researcher to systematically identify recurring themes and relationships across the cases studied. By reaching closure, the research ensures that the findings are robust, well-supported by the data, and ready for presentation in the final conclusions.

4. Results and discussion

This study highlights the distinct characteristics of fashion and furniture MSMEs, both of which represent subsectors of Indonesia's creative economy. While both sectors exhibit dynamic capabilities in their omnichannel transformation, their approaches differ significantly due to their unique operational and market demands. Fashion MSMEs focus heavily on real-time inventory synchronization and customer engagement. For example, the use of Instagram and Shopee not only facilitates product promotions but also enables personalized marketing strategies to attract their target demographic. This reflects their reliance on sensing and seizing capabilities to anticipate and meet customer preferences. Leadership in fashion MSMEs plays a pivotal role in aligning digital tools with business goals, fostering an *open innovation culture* that encourages experimentation and rapid adaptation to market changes. Furniture MSMEs, on the other hand, prioritize logistical efficiency and material procurement. Their operations are characterized by a pragmatic approach to supply chain coordination, with collaborative alliances enabling access to critical resources for production. For instance, partnerships with local suppliers enhance their ability to meet bulk orders while maintaining flexibility in customization. This demonstrates their transforming capabilities in addressing operational challenges, supported by a culture of gradual technological adoption rather than radical innovation.

4.1. The stages of omnichannel transformation in MSMEs

The implementation of omnichannel strategies has become crucial for MSMEs, not only in maintaining brand consistency across various platforms but also in enhancing operational efficiency and customer satisfaction, as well as strengthening the company's competitiveness in facing dynamic market challenges. Ultimately, this contributes to business success, reinforcing previous research by (Barbosa and Casais, 2022; Chen et al., 2019; Saghiri et al., 2017). MSMEs in this study stated that the most important aspect in implementing omnichannel is an integrated approach. Fashion MSMEs emphasized the need for real-time inventory updates and customer engagement across multiple platforms. As one fashion MSME owner noted:

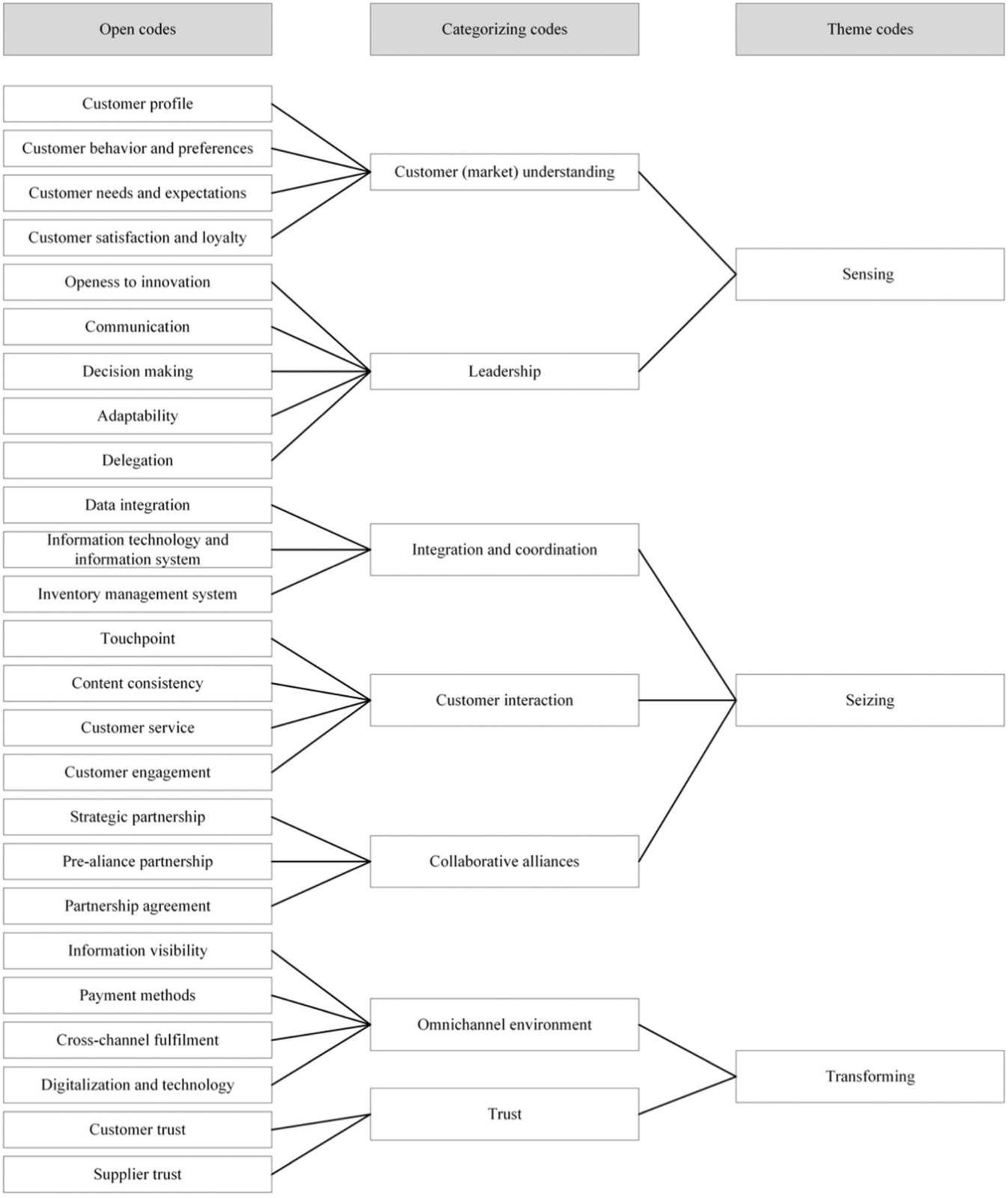


Fig. 3. Data structure based on content analysis.

We need to ensure customers see consistent stock availability across channels to avoid disappointment (Fashion MSMEs, Speaker 1).

Conversely, furniture MSMEs highlighted the importance of logistics and material procurement, with one owner stating:

Our priority is to ensure raw materials are available on time to meet customer demands for bulk orders (Furniture MSMEs, Speaker 1).

These differences underline the distinct operational focuses of each sector, shaped by their respective market demands. Fashion MSMEs underline the importance of consistent experience across all channels as

a basic form of brand presence. Furniture MSMEs state that omnichannel offers convenience for customers during the purchasing process.

The MSMEs participating in this study followed the stages of the evolutionary approach in omnichannel implementation as proposed in the literature (Cao and Li, 2015), but at a different pace. The transformation towards omnichannel in both MSMEs refers to the framework shown in Fig. 1. Both MSMEs stated that the existence of multiple channels is the starting point for implementing omnichannel, which corresponds to stage 1 of the framework. To create a consistent experience for clients, both MSMEs presented channel integration as an achievement.

4.1.1. Multichannel (Silo mode)

In this study, both MSMEs did not go through a single channel phase. The first stage fashion and furniture MSMEs, which began operations in 2013, involved using multiple sales channels, despite managing them separately. Initially, fashion MSMEs focused on a very limited sales channel. They operated as resellers, sourcing finished goods and selling them directly to customers.

> The initial reason was that I was not familiar with branding or marketing yet. Initially, the focus was solely on procuring the products and then selling them as such (Fashion MSMEs, Speaker 1).

They concentrate solely on selling goods and utilize Facebook for promotion. For offline sales, they convert their residence into a physical store for offline store, operate on car-free days, and participate in local bazaars. On the other hand, furniture MSMEs launch their businesses with a blend of offline and online strategies, selling through social media and displaying their products at other retail outlets.

> When we began, it seemed as if we were selling both online and offline. We began transporting stools to nearby shops while selling via Instagram (Furniture MSMEs, Speaker 1).

At this stage, MSMEs start using two or more platforms for sales and promotions, but with separate operations and lack of integration. Despite the opportunities they create through their presence on various channels, they still face a number of obstacles. Both MSMEs face challenges in maintaining brand consistency and marketing messages across multiple channels. It could potentially lead to confusion among customers and a decrease in loyalty. Moreover, customers are usually required to select a single channel for their purchases, which results in a decrease in overall flexibility and convenience.

4.1.2. Multichannel level 1 (minimum integration)

As time passes, fashion MSMEs have expanded their online sales channels through websites, Instagram, LINE, WhatsApp, and marketplaces. They have switched from using their residences as physical stores to renting a store instead. Prices are set at the same level for all sales channels. Fashion MSMEs have altered their business processes by producing their own products and diversifying. The shift is caused by suppliers' uncertainty about the availability of goods.

> Our suppliers have become unreliable; they sometimes have stock but not always. We opted to sew the products ourselves and looked for a home tailor. At the beginning, we only sewn plain t-shirts and sold them to students. We came across a tailor who was able to make pants, so we began making them and selling them as well (Fashion MSMEs, Speaker 1).

At this multichannel level-1 stage, fashion MSMEs have started integrating some basic functions between their physical stores and online sales channel using a system that helps manage inventory in a more integrated manner, although it is not yet fully automated. Their website can display the availability of stock in their physical stores thanks to their inventory management system.

> Through the new inventory management system, we can now show customers the stock available in our physical stores directly from the website (Fashion MSMEs, Speaker 2).

Furniture MSMEs utilize a variety of online sales channels, including Shopee, Tokopedia, Instagram, and WhatsApp. Prices are set at the same level across all sales channels. Some administrative tasks are still performed manually. The supervisor manually enters sales reports from both physical and online stores using a local information system. Meanwhile, inventory monitoring is conducted using spreadsheets that can be accessed by the supervisor, store employees, and the owner.

> Yes, the supervisor will coordinate with the employees who are responsible for the finishing process. The inventory monitoring spreadsheet will be updated every afternoon, and the shopkeeper can also observe the inventory level (Furniture MSMEs, Speaker 1).

> If there are 7 in stock, it means that there are 7 products available in total, I enter that amount in each marketplace. Later, if any are sold, I will update again in the marketplace and in the spreadsheet (Furniture MSMEs, Speaker 3).

Managing customer complaints is one of the challenges encountered at this stage. The reason for this is the lack of employees specifically designated to handle customer complaints.

> Now the supervisor has the most work. Managing social media, the marketplace and updating stock. What used to be done manually can now be done with an application. Including answering chats (Furniture MSMEs, Speaker 2)

> I also reply to consumer chats regarding stock, customization, shipping, and prices across all sales channels. Later, when it's busy, for example a flash sale, the co-owner will help to ensure all channels are served (Furniture MSMEs, Speaker 3).

Furthermore, the absence of real-time data integration leads to errors in stock updates and order management. By understanding these activities and challenges, fashion and furniture MSMEs can strive to optimize the integration of their sales channels, improve team coordination, and leverage technology to support more efficient operations. Based on the interviews, it is evident that furniture MSMEs remain at the multichannel level-1 stage (minimum integration) due to limited resources, inadequate technological infrastructure, and the pragmatic nature of the owners. Business owners who are like this tend to prioritize stability and business continuity over ambitious growth or radical innovation. Their primary goal is to ensure that operations go smoothly, employees are paid, and a small profit is left for business security.

4.1.3. Multichannel level 2 (moderate integration)

Fashion MSMEs do not go through these stages. They transitioned from multi-channel level-1 to omnichannel using the appropriate level of resources and commitment. These findings are in line with previous research that suggests companies can skip the intermediate stages in certain circumstances (Berman and Thelen, 2016; Simone and Sabbadin, 2018).

4.1.4. Omnichannel (full integration)

At this omnichannel stage, fashion MSMEs have facilitated click-and-pick-up in-store and buy-online-and-return-in-store services. To ensure consistent availability of goods, it is necessary to better coordinate and synchronize warehouse and store inventory.

> Customers have the option to order our products online through Marketplace or WhatsApp, and then pick them up at our shop. To avoid customer disappointment when ordered goods are not available, warehouse inventory must always be in sync with the marketplace (Fashion MSMEs, Speaker 1).

Customers can pick up their orders from the store or return online purchases to the physical store seamlessly (Fashion MSMEs, Speaker 2).

The fashion MSMEs use Instagram as the main media to promote new products and direct customers to their online sales channels, so that customers can buy through the sales channels that best suit them.

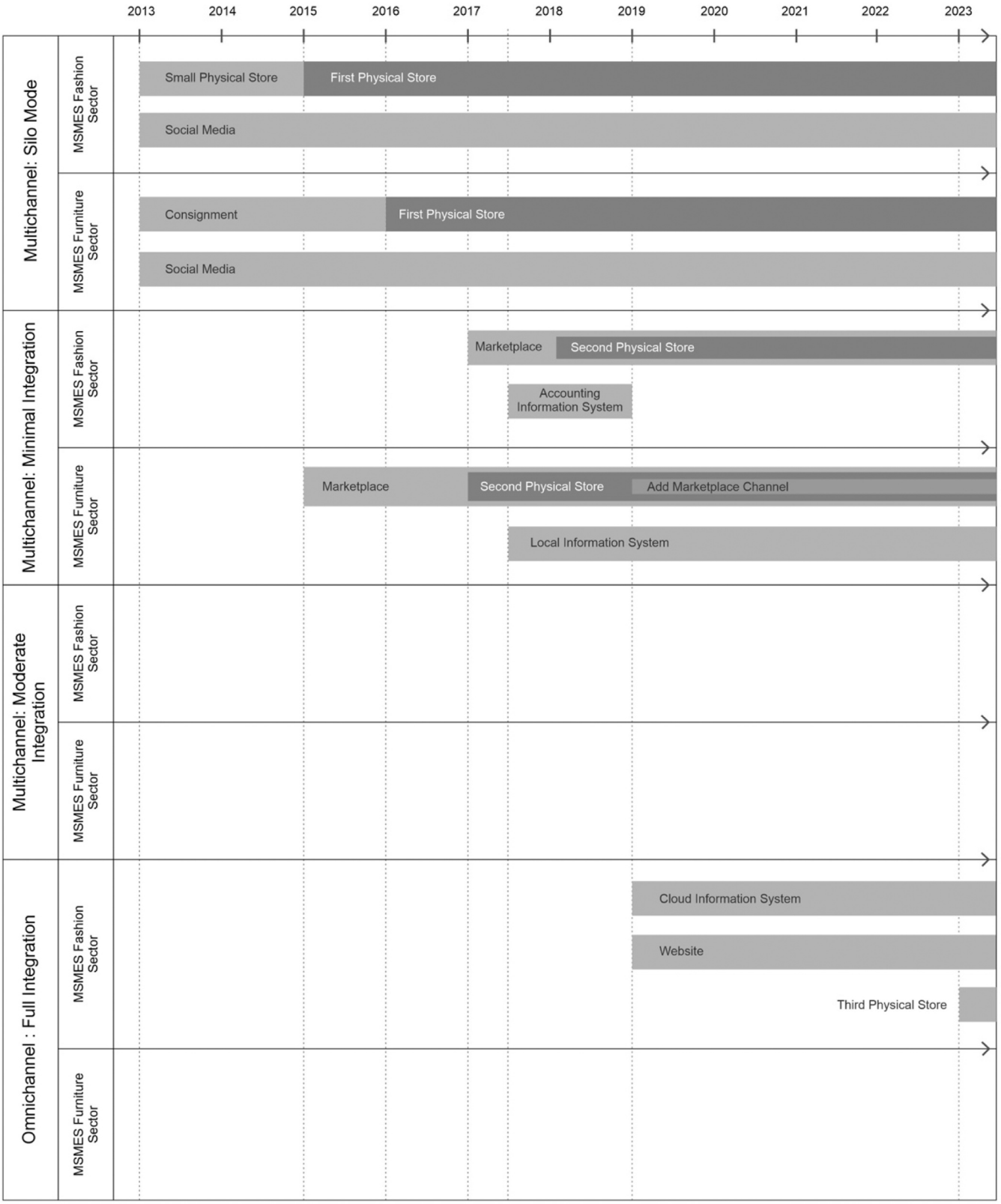


Fig. 4. Transformation stages towards omnichannel in fashion and furniture MSMEs.

> Our main store, aside from physical locations, is Instagram. Whenever we unveil a new product on Shopee, we make sure to announce it first on Instagram (Fashion MSMEs, Speaker 1).

Fashion MSMEs utilize a cloud-based information system to handle sales, inventory, and business operations. This platform allows them to manage their business more efficiently and effectively, with a user-friendly interface and various features that can be customized to meet the specific needs of each business:

> We subscribed to I-Seller for 10 years because if we did it every month, it would be more expensive. I-Seller is utilized to cover 8 warehouses, 5 stores, and all online channels (Fashion MSMEs, Speaker 1).

> Customers who visit the store will receive a voucher to use later on the website (Fashion MSMEs, Speaker 1).

Furniture MSMEs have not yet achieved full omnichannel integration due to limited resources and technological infrastructure. Operational processes such as inventory monitoring and sales reporting are still done manually. The lack of real-time data integration leads to stock inaccuracies and challenges in team coordination. The primary focus of these MSMEs is on maintaining operational stability and business continuity, delaying the priority for full omnichannel transformation.

Even after system integration, the challenge of real-time data and inventory synchronization persists, specifically in maintaining consistency across different sales channels. Managing operations across multiple warehouses and physical stores, as well as various online channels, can also be a challenge due to complex coordination and logistical issues. Additionally, maintaining a consistent customer experience across all channels, both online and offline, requires continuous efforts in staff training and service quality monitoring. Furthermore, implementing new technology and making major investments in omnichannel systems necessitate careful planning and budgeting. Fashion MSMEs can optimize their strategies for implementing omnichannel efforts by understanding these challenges and improving operational efficiency and customer experience across all channels. The transformation stages undergone by fashion and furniture MSMEs are shown in Fig. 4.

4.2. Capabilities developed by MSMEs in transformation towards omnichannel implementation

Internal and external factors play a critical role in driving the transformation of MSMEs towards omnichannel strategies. Internally, factors such as managerial commitment are crucial, as leadership must be dedicated to investing in and supporting the transition. The availability of resources, including financial, technological, and human capital, significantly influences how quickly and effectively MSMEs can integrate multiple channels. Externally, technological advancements push MSMEs to adopt new systems and platforms to remain competitive, while market and consumer behavior changes, such as the demand for seamless shopping experiences, compel MSMEs to implement omnichannel strategies. Lastly, competition pressures MSMEs to innovate and adopt these strategies to maintain their market position and attract customers. Together, these internal and external factors form the foundation for successful omnichannel transformation in MSMEs.

The transformation process toward omnichannel in MSMEs involves both ordinary and dynamic capabilities, as identified in prior studies (Mrutzek-Hartmann et al., 2020; Solem et al., 2023). Ordinary capabilities ensure efficient daily operations, such as production planning, raw material procurement, and human resource management. Meanwhile, dynamic capabilities, categorized into sensing, seizing, and transforming, enable MSMEs to adapt to changing market conditions and customer demands. Specifically, from Mrutzek-Hartmann et al. (2020), five dynamic capabilities were identified: customer (market) understanding, integration and coordination, customer interaction, omnichannel environment, and innovativeness. This research further adds collaborative alliances, leadership, and trust as dynamic capabilities observed through case study analysis. Collaborative alliances provide access to critical resources and foster innovation. Strong leadership ensures strategic alignment and effective decision-making, while trust facilitates secure transactions and enhances both internal collaboration and external relationships.

This study will categorize the eight capabilities into sensing, seizing, and transforming, in contrast to prior studies (Mrutzek-Hartmann et al., 2022a,b) that classified dynamic capabilities into frontend and backend activities. The purpose of grouping capabilities into sensing, seizing, and transforming is to understand how an organization detects opportunities, takes advantage of those opportunities, and adjusts resources to adapt to changes (transforming). This approach helps analyze the MSMEs response to market and technological changes in a more structured way.

4.2.1. Sensing

According to the results of the empirical study, the sensing can be categorized into customer (market) understanding and leadership. Customer (market understanding) capabilities involves identifying and interpreting changes in the market and customer preferences. This involves gathering market insights and analyzing customer behavior to guide strategic decisions, as one interviewee stated:

> I've done a survey to open an offline store yesterday, I've tried to make a cost-benefit analysis, the competitors there, how is the visitor traffic, I've started the survey but haven't executed it yet, the study is still shallow, still not in-depth enough (Fashion MSMEs, Speaker 1).

Another interviewee highlighted the importance of targeting specific demographics to shape offerings:

> Our main target market is women aged 18–30, so we sell more basic items. For example, trousers made of drill material, elastic waistbands to attract college students, and simple shirts made of comfortable materials. But in reality, many of our customers are over 30, although the number is lower than the 18–30 age group (Fashion MSMEs, Speaker 2).

In addition, respondents from the furniture MSMEs emphasized their understanding of customer demands:

> The sales data shows that the majority of our consumers are women between the ages of 20 and 35. Young couples who are starting to furnish their homes. We can even ask for items that have already been purchased and sent when the house is finished (Furniture MSMEs, Speaker 2).

Leadership plays a critical role in sensing emerging trends and making strategic decisions based on these market insights. Leaders guide the organization in interpreting market signals and aligning resources effectively. As one interviewee noted:

> I look for the right technology. So I think this digitalization starts with the people, not the tools. Tools are just accelerators so that the business process is neater, monitoring is faster, but if the goal is not achieved, it means nothing, right? So I try to align technological tools with human capability and business goals (Fashion MSMEs, Speaker 1).

Another respondent shared their proactive approach to adapting and learning:

> If the strategy is still the same as before but hoping for something new, it's impossible. So from now on I'm learning, how can I keep up with technology. I'm learning how to increase online followers which can later increase turnover. I want to understand how to treat a good online business (Fashion MSMEs, Speaker 1).

Sensing capabilities, including customer (market) understanding and leadership, form the foundation for effective omnichannel

transformation. By leveraging these capabilities, MSMEs can proactively identify market opportunities, understand customer preferences, and strategically position themselves to adapt to dynamic changes in their industries. These insights underscore the importance of continuous learning, strategic decision-making, and aligning technological advancements with organizational goals.

4.2.2. Seizing

Once these opportunities are identified, seizing encompass the allocation of resources, strategic decision-making, and the reconfiguration of assets to effectively exploit these opportunities and gain a competitive edge. This study identified three seizing capabilities, including: integration and coordination, customer interaction, and collaborative alliances. Integration and coordination across channels and departments are essential to effectively capitalize on market opportunities. Fashion MSMEs, for example, emphasized the importance of their inventory management system:

> Through the new inventory management system, we can now show customers the stock available in our physical stores directly from the website" (Fashion MSMEs, Speaker 2).

Similarly, another participant highlighted the challenges of implementing a system while prioritizing financial stability:

> I taught and accompanied the supervisor to start using the system, but it did not work. Then I gave up. What is now important is that the sales keep increasing so we can pay employee salaries and store rent" (Furniture MSMEs, Speaker 1).

Customer interaction is another vital capability for seizing opportunities. Businesses use digital tools and communication strategies to enhance customer experiences. For instance, Fashion MSMEs utilize synchronized systems to ensure inventory accuracy and prevent customer dissatisfaction.

> Customers have the option to order our products online through Marketplace or WhatsApp, and then pick them up at our shop. To avoid customer disappointment when ordered goods are not available, warehouse inventory must always be in sync with the marketplace" (Fashion MSMEs, Speaker 1).

Additionally, participants highlighted the role of social media platforms in driving customer engagement.

> Our main store, aside from physical locations, is Instagram. Whenever we unveil a new product on Shopee, we make sure to announce it first on Instagram (Fashion MSMEs, Speaker 1).

Collaborative alliances enable MSMEs to access resources, share expertise, and expand their market reach. Furniture MSMEs, for instance, emphasized the importance of partnerships.

> We often collaborate with local businesses to create exclusive products. This helps us reach a broader customer base and share resources, such as promotional campaigns (Fashion MSMEs, Speaker 2).

4.2.3. Transforming

These capabilities include omnichannel environment, innovativeness, and trust. In an omnichannel environment, this means regularly updating operations, technology, and strategies to align with changing market demands. It also includes fostering innovation through new products or business models and building trust with stakeholders, which is essential for sustaining long-term transformation and successful collaborations. In this phase, businesses integrate and upgrade their operational systems to align with evolving market demands. For example, a Fashion MSME explained how they adopted a comprehensive system to manage multiple sales channels:

> We subscribed to I-Seller for 10 years because if we did it every month, it would be more expensive. I-Seller is utilized to cover 8 warehouses, 5 stores, and all online channels (Fashion MSMEs, Speaker 1).

Innovation plays a critical role in transforming operations, enabling businesses to introduce new products or adapt existing ones to meet market trends. A furniture MSME noted:

> We try to develop products that cater to modern and minimalist concepts because that's what the younger market seems to prefer these days (Furniture MSMEs, Speaker 2).

Building trust with stakeholders, including customers, partners, and team members, is vital for sustaining long-term transformation. A fashion MSME highlighted the importance of trust in maintaining strong customer relationships:

> I want to maintain transparent communication with customers, ensuring they trust our brand and the services we offer, even across different platforms" (Fashion MSMEs, Speaker 1).

These transforming capabilities enable MSMEs to adapt effectively to the omnichannel landscape, ensuring their operations remain relevant and competitive. By fostering innovation, maintaining trust, and embracing omnichannel integration, MSMEs position themselves for sustained success in a rapidly evolving market environment.

The competitive advantages that MSMEs gain after implementing omnichannel strategies include several key benefits. First, enhanced customer experience is achieved by providing a seamless and consistent interaction across all customer touchpoints, both online and offline, leading to improved customer satisfaction, engagement, and ultimately, loyalty and sales. Additionally, improved operational efficiency results from the integration of digital systems and better coordination across sales channels, allowing MSMEs to manage inventory more effectively, reduce order processing errors, and streamline their operations. Furthermore, omnichannel strategies facilitate a broader market reach by enabling MSMEs to tap into a wider audience through multiple sales channels, including e-commerce platforms, social media, and physical stores. Lastly, the implementation of omnichannel approaches strengthens brand consistency by ensuring uniform branding and marketing messages across all channels, minimizing customer confusion and enhancing brand recognition. Referring to the framework in Fig. 2, the dynamic capabilities involved in the transformation process towards omnichannel in fashion and furniture MSMEs are illustrated in Fig. 5.

Fashion MSMEs focus on digital system integration and neat supply chain management to maintain quality and production efficiency. Meanwhile, furniture MSMEs place a high priority on flexibility and adaptability when it comes to producing and procuring raw materials. Despite the presence of numerous manual processes, technology is steadily improving efficiency. Both MSMEs possess good operational management to aid in the transformation towards omnichannel, with each strategy customized to their needs and capacities. The human resource management of furniture and fashion MSMEs differs. Furniture MSMEs have a stronger reliance on experience and direct learning at work, while fashion MSMEs rely on technology and more structured systems to improve efficiency. Both are trying to overcome internal coordination challenges with the help of employee competencies and technology implementation. Both prioritize communication and customer service as part of their human resource management strategies.

The most significant capability categories were determined by analyzing how frequently they were cited in interview transcripts and other supporting data for each MSME. To assess the various categories, the frequency of mentions was used as a weighted indicator. Fig. 6 shows the overall coding results of data analysis. The total coding results in percentages are a representation of the total coding for each capability category in relation to the total number of codes. The next two columns

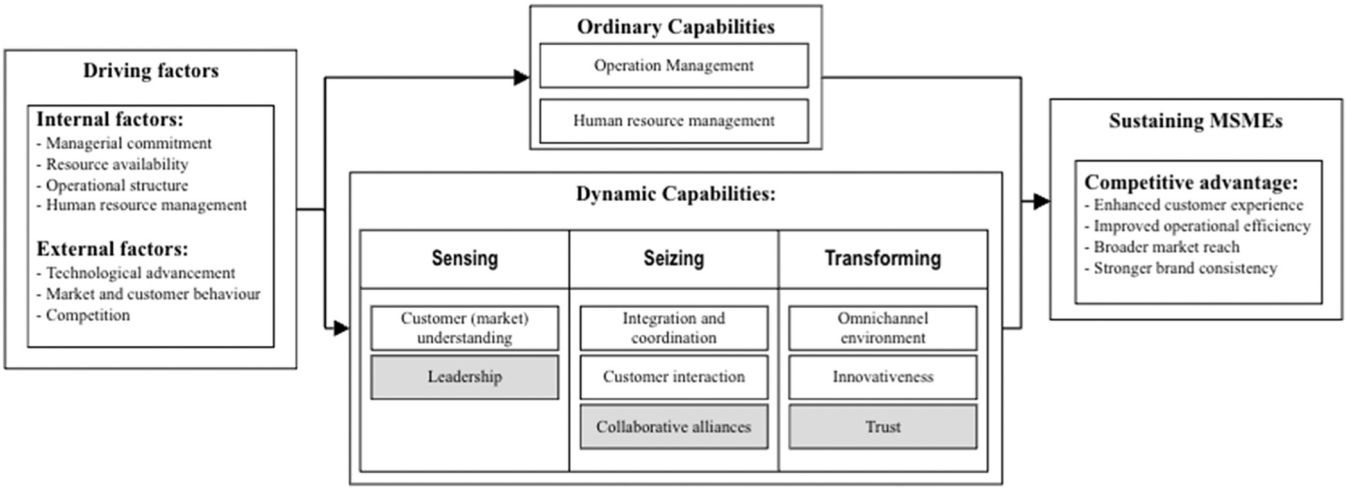


Fig. 5. Dynamic capabilities framework for transformation towards omnichannel in MSMEs.

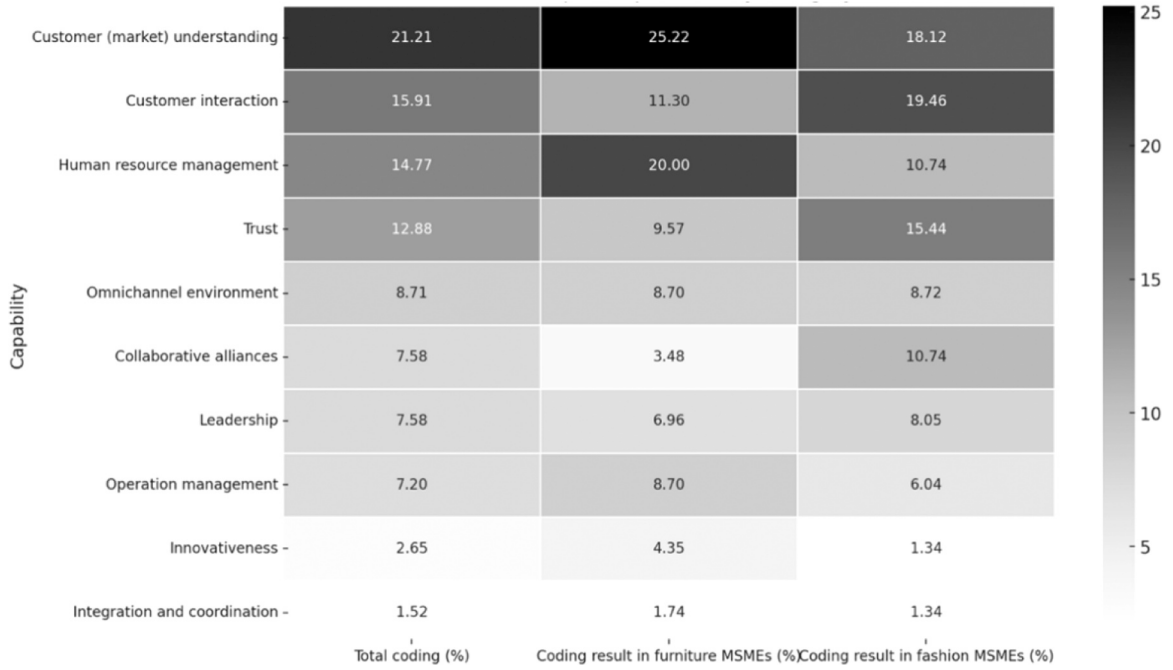


Fig. 6. Heatmap of capabilities by category.

show results per MSME. The color intensity in the heatmap visually represents these percentages, where darker shades indicate higher frequencies of citations, thus signifying greater importance.

Fig. 6 provides several important insights into how capabilities are distributed across total coding, furniture MSME, and fashion MSME. Customer (market) understanding emerged as the most significant capability, with a notably higher importance in furniture MSMEs. In contrast, innovativeness and integration and coordination were the least emphasized capabilities, indicating potential areas for development or reduced relevance in both sectors. Fashion MSMEs placed greater emphasis on customer interaction and trust, reflecting their strategic focus on enhancing customer engagement and fostering reliability. Conversely, furniture MSMEs prioritized human resource management and innovation, showcasing their emphasis on internal resource optimization and creative adaptation.

Interestingly, the omnichannel environment scored consistently across both MSMEs, underlining its universal significance in achieving seamless channel integration. The heatmap thus provides a comprehensive visual representation of the relative importance of dynamic capabilities, highlighting sector-specific strengths and weaknesses. These findings offer actionable insights into where each MSME can allocate resources and refine strategies to optimize their omnichannel transformation journey.

4.3. Relevance of dynamic capabilities and open innovation framework

The findings of this study highlight the critical interplay between dynamic capabilities—sensing, seizing, and transforming—and *open innovation dynamics* in driving MSME transformations towards omnichannel strategies. This interplay is evident in how MSMEs in fashion and furniture sectors leverage both internal capabilities and external networks to overcome resource constraints and meet evolving market demands.

Collaborative alliances play a pivotal role in addressing the bounded rationality of MSMEs (Chesbrough and Bogers, 2014) by enabling access to external resources and expertise. For furniture MSMEs, partnerships

with local suppliers and logistics providers enhance their ability to meet customer demands for bulk orders while maintaining flexibility in operations. These alliances exemplify Chesbrough (2003) open innovation framework, where external collaboration fosters innovation and operational resilience. By engaging in collaborative alliances, MSMEs can overcome structural and operational limitations, ensuring competitiveness in a resource-constrained environment.

Leadership emerged as a key enabler of open innovation culture within MSMEs. In fashion MSMEs, leaders play a critical role in fostering a culture of experimentation and continuous learning. This involves aligning technological tools with business objectives to enhance customer engagement and streamline operations. For instance, the adoption of digital tools like inventory management systems reflects how leadership integrates both internal innovation processes and external market opportunities. This aligns with (Chesbrough and Bogers, 2014), who emphasize the importance of managing open innovation in small firms to sustain competitive advantage.

The application of open innovation dynamics within the creative economy showcases the adaptive strategies employed by MSMEs to address sector-specific challenges. Fashion MSMEs focus on customer-centric innovations, such as personalized marketing and synchronized online-offline operations. In contrast, furniture MSMEs prioritize logistical efficiency and product customization through external partnerships. These sector-specific adaptations demonstrate how MSMEs operationalize dynamic capabilities within the broader framework of open innovation.

By integrating dynamic capabilities and open innovation dynamics, this study provides actionable insights for MSMEs navigating omnichannel transformation. Sensing capabilities allow MSMEs to identify market opportunities and emerging trends, while seizing and transforming capabilities enable them to capitalize on these opportunities through strategic resource allocation and operational adjustments. The incorporation of open innovation dynamics further amplifies these capabilities, enabling MSMEs to balance internal efficiency with external collaboration effectively.

4.4. Theoretical implications

The findings align with the works of Teece (2007) who emphasized that dynamic capabilities are vital for firms operating in rapidly changing environments, allowing them to adapt, innovate, and maintain competitiveness. In the context of omnichannel strategies, the study reinforces the argument that these capabilities are essential for businesses to integrate diverse channels and provide a seamless customer experience, as suggested by (Barbosa and Casais, 2022). The study also supports the idea that not all firms follow a linear progression in their omnichannel journey, as companies with stronger dynamic capabilities can skip certain stages of integration, as discussed by Berman and Thelen (2016). However, this research contributes to the literature by addressing the unique context of MSMEs, particularly in resource-constrained environments, such as those in the furniture and fashion MSMEs. This aligns with the critical viewpoints of Eisenhardt and Martin (2000) who argue that dynamic capabilities do not inherently ensure a competitive edge but are contingent upon the specific context in which they are applied. The study extends these critiques by showing that furniture MSMEs, despite demonstrating dynamic capabilities such as sensing and seizing opportunities, face slower omnichannel adoption due to pragmatic management approaches and resource constraints. Conversely, fashion MSMEs exhibit more agility in leveraging these capabilities, reflecting how sectoral differences shape the application of dynamic capabilities. By integrating these sector-specific insights, this study expands the theoretical understanding of dynamic capabilities in the omnichannel context, emphasizing their conditional effectiveness and the need for tailored strategies in MSMEs.

4.5. Managerial implications

The findings of this study provide actionable insights for MSME managers, particularly in the fashion and furniture sectors, emphasizing the importance of integrating dynamic capabilities with open innovation dynamics to facilitate omnichannel transformation. Managers should focus on fostering adaptive strategies that leverage both internal capabilities and external networks to address resource constraints and respond effectively to market demands. For fashion MSMEs, dynamic capabilities such as sensing market trends and seizing customer preferences can be used to develop personalized marketing campaigns and implement real-time inventory management systems, ensuring consistent stock availability across channels and reducing missed sales opportunities. Furniture MSMEs, on the other hand, can prioritize transformative capabilities by collaborating with external partners, such as logistics providers and suppliers, to drive innovation in product customization and optimize supply chain management.

Collaborative alliances, as highlighted in this study, play a critical role in enabling MSMEs to access external resources and expertise, allowing them to overcome operational and structural constraints. Managers should actively seek partnerships that can complement their internal capabilities and enhance operational resilience, particularly in resource-constrained environments. For instance, engaging with local suppliers can help furniture MSMEs meet customer demands for bulk orders while maintaining flexibility. Leadership is another pivotal factor in embedding an open innovation culture within MSMEs. Managers should cultivate a culture of experimentation and continuous learning, aligning technological tools with business objectives to enhance customer engagement and streamline operations. For example, adopting digital solutions, such as inventory management systems, can help integrate internal innovation processes with external opportunities, ultimately sustaining competitiveness. By integrating dynamic capabilities—sensing, seizing, and transforming—with open innovation dynamics, managers can align their strategies to balance internal efficiency with external collaboration. This alignment not only addresses resource constraints but also ensures that MSMEs remain agile and responsive to the dynamic demands of the omnichannel marketplace.

5. Conclusion

This study explored the stages of omnichannel transformation and the dynamic capabilities required by MSMEs, focusing on two distinct cases: a fashion MSME and a furniture MSME. The findings highlight that transformation processes are deeply influenced by the unique characteristics, operational priorities, and resource availability of each company rather than by sectoral generalizations. Fashion MSME revealed a focus on leveraging digital tools for seamless integration across channels, prioritizing real-time inventory management and consistent customer engagement. These practices underscore their reliance on dynamic capabilities such as sensing market trends and seizing opportunities through enhanced customer interaction. In contrast, furniture MSME emphasized logistics optimization to meet bulk customer demands, showcasing capabilities like transforming operations and fostering collaborative partnerships. Their pragmatic approach, centered on stability and resource conservation, reflects a slower but steady progression in omnichannel adoption. These differences demonstrate that omnichannel transformation is not a one-size-fits-all process. Rather, it is shaped by the specific challenges and opportunities faced by each business. By focusing on individual company characteristics, this study contributes to a nuanced understanding of how MSMEs navigate omnichannel integration, providing actionable insights into tailoring strategies for diverse contexts.

This study is based on two MSME case studies—one from the fashion MSME and one from the furniture MSME—which limits its generalizability. The conclusions drawn are deeply rooted in the specific operational and market contexts of these businesses, potentially overlooking

broader patterns present in other sectors or regions. Additionally, the focus on internal factors, such as resource allocation and managerial priorities, may not fully capture the impact of external influences like market competition or technological ecosystems. To address these limitations, future research should aim to include a broader and more diverse range of case studies. Expanding the sample size would enable researchers to extract generalizable insights while maintaining the ability to draw sector-specific and context-sensitive conclusions. Furthermore, integrating external factors into the analysis would provide a more holistic understanding of the omnichannel transformation process. Future research should explore a wider range of MSMEs across industries, sizes, and regions to enhance cross-case comparisons and generalizability. Longitudinal studies could provide deeper insights into the stages of omnichannel transformation, while examining external factors like policies, technology ecosystems, and market competition would offer a more comprehensive understanding of enablers and barriers.

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Ethical statement

This study and its instruments have received the ethical approval from Universitas Gadjah Mada (UGM) Research Ethics Commission No. KE/UGM/129/EC/2024

CRediT authorship contribution statement

Febriani Atik: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Data curation. **Sopha Bertha Maya:** Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Wibisono M. Arif:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Apendix 1. : Interview Guideline

Dynamic capabilities for omnichannel transformation in MSMEs: A comparative case study of fashion and furniture sectors

Introduction

1. Begin by expressing gratitude to the interviewee for their participation in this significant project.
2. Notify the interviewee that the session will be recorded and transcribed.
3. Provide a brief description of the research and its objectives.
4. Collect the following details for potential follow-up and methodological accuracy:

a) Interviewee's phone number
b) Date of the interview
c) Duration of the interview
d) Location of the interview

General Questions about the Interviewee

1. Can you please tell me your name?
2. How long have you been working here?
3. What division are you part of, and what are your responsibilities?

Part A: General information about the business

1. Main products and services
2. Product characteristics
3. Target market
4. Year the business started
5. Business location
6. Business licenses
7. Capital and budget allocation
8. Number of employees and their duties
9. Sales channels used
10. Pricing strategy
11. Strategic approach used (Cost-Efficient or Responsiveness)
12. Annual turnover

Part B: Information about routine activities and governance in business

1. HR Management (status of employees: permanent/daily/contract-based, payroll methods, benefits received)
2. Relationships with Suppliers
3. Customer Relations
4. Shipping, Loading-Unloading, Transportation Modes Used
5. Marketing Activities
6. Risk Management

Part C: Information about omnichannel

1. Knowledge about omnichannel and initial interest in omnichannel.
2. Advantages and disadvantages of omnichannel implementation.
3. Service planning developed in omnichannel implementation.
4. Evaluation of current omnichannel implementation.

Part D: Transformation process

1. Describe the process of transitioning from multichannel to omnichannel (the specific steps involved in this transformation, such as restructuring operations, training staff, or adopting new technologies, and the timeline for these changes)
2. Explain which new sales channels (e.g., online stores, mobile apps) were developed as part of the omnichannel implementation.
3. Identify any services or sales channels that remain unintegrated into the omnichannel strategy and the reasons for this.
4. The digital enhancements made to physical stores, such as providing Wi-Fi, using tablets for customer service, or integrating online store access.
5. Future plans for further digitization and technological integration at physical points of sale.
6. Any changes or upgrades made to storage facilities and warehouse management systems to accommodate the demands of an omnichannel operation, such as inventory management software or restructured warehouse layouts.
7. The challenges encountered during the omnichannel transformation, such as integration issues, technological gaps, or resistance to change. Explain how these challenges were addressed and whether they were anticipated before the transformation began.
8. The modifications made to customer interaction and communication strategies across various channels, ensuring a consistent experience.

9. Indicate the timeline for establishing a social media presence, the platforms used, and how your offerings have evolved in response to social media engagement.
10. The tools and methods used to analyze customer data, such as social media analytics, and how these insights have influenced business decisions and changes.
11. Identify any external assistance, such as consultants, technology providers, or partnerships, that supported the transformation process.
12. The evolution of pricing strategies across sales channels, including whether pricing has been standardized or differentiated, and the rationale behind these decisions.
13. The methods used to ensure consistent product availability across various sales channels, such as inventory management systems or logistical coordination.
14. Detail the strategies and forecasting techniques employed to manage inventory effectively, including any adjustments made to improve accuracy or efficiency.
15. The relationship with logistics partners, including joint projects or collaborations to enhance goods procurement and delivery.
16. Reward systems, such as discount or loyalty programs.
17. Innovation management practices and infrastructure have been adapted or restructured to support the omnichannel transformation.
18. The payment systems, particularly contactless options, that have been adopted or updated to meet customer expectations and enhance the shopping experience across channels.
19. Summarize from Interviewee perspective: What are the key factors for success in omni-channel retailing? What are the essential components for this strategy?

Closing

1. Thank the interviewee for their time and contributions, acknowledging the value of their insights.

Inform the interviewee about what will happen next, such as how their responses will be used, the timeline for the research, and requesting willingness to be contacted again, for necessary follow-up or to share results

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